

Generation Interconnection System Impact Study Report

For Queue Project Q12-0168

Cmc 138 kV
October 01, 2013

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q12-0168. The Interconnection Customer proposes a 256.6 MW wind facility interconnecting at the Cmc 138 kV substation in the Pennsylvania area. The request was received on October 27, 2012. The study evaluated system intact and single contingency conditions using the regional planning case, including 131 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 8 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$26,521,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of Ft Humbug / Orchard Road 500kV, the Ft Humbug 500/345 kV Auto loads to 128.1% of its 628 MVA rating. Project DFAX: 6.0%.

For the loss of Claremont And Montano Tap / Cmc 138kV, the Colver Power Project / Cmc 138kV loads to 122.6% of its 207 MVA rating. Project DFAX: 100.0%.

The Nottely / Marion S 138kV is overloaded to 118.4% of its 209 MVA emergency rating for the loss of the Justin Lane / Rollins Wind Project 138kV L. The project contributes 5.9% DFAX (15.1 MW) to this overload.

The Claremont And Montano Tap / Cmc 138kV is overloaded to 117.8% of its 215 MVA emergency rating for the loss of the Colver Power Project / Cmc 138kV. The project contributes 100.0% DFAX (256.6 MW) to this overload.

For the loss of Rollins Wind Project Substation / Justin Lane 138kV, the 7Th Avenue N / Nottely 138kV loads to 117.4% of its 213 MVA rating. Project DFAX: 5.9%.

The North Allegheny Windpower Project / Wahiawa 345kV is overloaded to 103.8% of its 1,224 MVA emergency rating for the loss of the Sunrise Ctr / Merret 500kV. The project contributes 10.6% DFAX (27.3 MW) to this overload.

For the loss of James E. Rogers Energy Complex / Collierville 138kV, the N Allegheny Windpower Project 345-138 kV Transformer loads to 103.7% of its 365 MVA rating. Project DFAX: 6.6%.

The Johnstown / Claremont And Montano Tap 138kV L is overloaded to 102.4% of its 234 MVA emergency rating for the loss of the Cmc / Colver Power Project 138kV Line. The project contributes 100.0% DFAX (256.6 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Ft Humbug 500/345 kV Auto	Replace transformer with higher-rated unit	\$29,359,000	13% (\$3,766,000)
Colver Power Project / Cmc 138kV	Rebuild line with higher-capacity conductor	\$3,521,000	100% (\$3,521,000)
Nottely / Marion S 138kV	Rebuild line with higher-capacity conductor	\$5,186,000	40% (\$2,062,000)
Claremont And Montano Tap / Cmc 138kV	Rebuild line with higher-capacity conductor	\$5,476,000	100% (\$5,476,000)
Nottely / 7Th Avenue N 138kV Line	Rebuild line with higher-capacity conductor	\$11,976,000	40% (\$4,761,000)
North Allegheny Windpower Project / Wahiawa 345kV	Reconductor line	\$9,349,000	14% (\$1,348,000)
N Allegheny Windpower Project 345-138 kV Transformer	Replace transformer with higher-rated unit	\$17,117,000	27% (\$4,696,000)
Johnstown / Claremont And Montano Tap 138kV L	Reconductor line	\$891,000	100% (\$891,000)
	Total allocated to project		\$26,521,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.